Markov State Modelling of the HP35 Domain
Final Project for the Course “Biological Datasets for Computational Physics”

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I. INTRODUCTION

The villin headpiece subdomain HP35 is one of the smallest proteins that folds autonomously, and one of the fastest. Its native fold is a compact three-helix bundle whose stability derives largely from a hydrophobic core formed by the aromatic residues *Phe6*, *Phe10* and *Phe17*. The Lys24-Nle/Lys29-Nle double mutant (Fig.1) studied here was experimentally found to fold in $\simeq 0.7\,\mu$s at 300 K with little temperature dependence [1]. Because of its short folding time, already in the early 2010s it has been possible to achieve long unbiased all-atom simulations that resolve dozens of folding and unfolding events [2], which made it become a paradigmatic model system for protein folding studies. A protein that folds this fast, over so low a barrier, is expected to populate only marginally stable intermediates that are difficult to resolve; yet a strict two-state description has never fully accounted for it. Both experimental [3] and simulation based studies [4–6] have reported evidence of structural heterogeneity within its conformational ensemble.

In this project, I analyse the well-known $305\,\mu$s trajectory of the Lys24-Nle/Lys29-Nle mutant published by Lindorff-Larsen *et al.* [2]. Thanks to the extensive sampling and the system's simplicity, the average folding time was estimated directly from the trajectory ($3.2 \pm 0.6\,\mu$s [9] - average waiting time in the unfolded state, which is $\times 3$ the experimental folding time). I build Markov State Models (MSMs) [7] to ask whether HP35 possesses resolvable metastable substructure (e.g. off-pathway traps, intermediates, substates within the folded basin, or partially folded states). Building MSM and here a difficulty arises that is not merely technical. The MSM pipeline (feature selection, dimensionality reduction, clustering) has no unique optimal recipe, and different reasonable choices can emphasise different slow processes. As a result, different pipelines may produce different sets of metastable states, with different populations. This motivates a deliberately redundant design: rather than committing to one pipeline, I build several and ask which conclusions survive across all of them and which are model dependent, if the description of the various model is compatible (i.e. at most one has more resolution and one less, but they need to agree at the coarsest level).

Concretely, four MSMs are constructed from two internal-coordinate representations (residue–residue contacts and backbone dihedral angles) each reduced by two methods, PCA and tICA. After spectral analysis and PCCA++ coarse-graining, the four models are compared on (i) the estimated folding timescale, (ii) the robustness of the resolved macrostates, and (iii) their structural interpretation.

II. METHODS

A. Dataset assembly

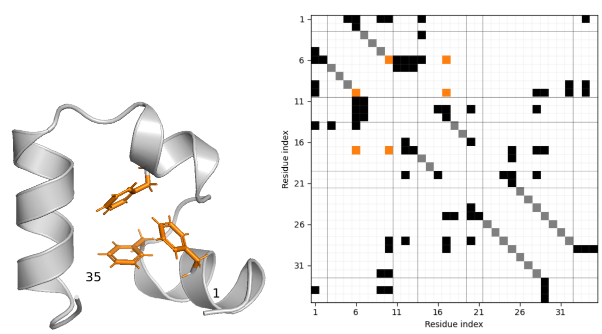

FIG. 1: ***Crystal structure and contact map of HP35*** *(PDB: 2F4K [1]). Left: The folded structure consists of three α-helices (3-10), (14-19), and (22-32), connected by two loops. Highlighted in orange are aromatic residues Phe6, Phe10, Phe17, whose interactions contribute to the formation of the hydrophobic core. Visualization with PyMOL [16]. Right: contact map computed with a distance cutoff of* $4.5\,A$*. Highlighted in grey are the intra-helix contacts* $(i,\, i+4)$*, and in orange the contacts between the three mentioned aromatic residues.*

Rather than the original all-atom trajectory of Lindorff-Larsen *et al.* [2], I use the preprocessed version publicly distributed by Nagel *et al.* [8], which provides internal-coordinate representations selected and filtered specifically for MSM construction (see [8] for feature definitions). Each time series contains $\simeq 1.5 \times 10^6$ frames at a sampling interval of $\simeq 200$ ps. I employ two of these representations as alternative feature sets: residue–residue contact distances (`hp35.mindists2`, 42 features) and backbone dihedral angles (`hp35.dihs.shifted`, 62 features). A third set of crystal-structure contact distances (`hp35.crystaldists`, 53 features) is used as an additional structural descriptor in later analysis. To enable structural visualization, I reconstructed the backbone atom coordinates from the raw dihedral-angle time series (`hp35.dihs`); all molecular renderings shown below are obtained this way.

B. Methods for MSM construction

All code is written in Python, using Deeptime [10] for MSM construction and validation and BioPython [11] for analysis of the crystal structure (PDB 2F4K). A Markov state model discretizes the continuous high-dimensional dynamics into a finite set of states with transitions observed at a fixed time interval, aiming for a reduced representation that preserves the slow dynamical processes of the original system. The pipeline used here consists of the following steps:

1. Feature selection (distances or dihedrals),
2. Dimensionality reduction (PCA or tICA),
3. Clustering of the reduced coordinates into discrete states (k-means),
4. Estimation of the transition matrix and selection of the Markov lag time τ_{MSM} (implied-timescale convergence test; see Supp. Methods B or [7, Ch. 2.4.1]),
5. Validation against the data (Chapman–Kolmogorov test; see [7, Ch. 4.8]).

The first two steps are the ones explored most in this work. Together they compress the raw MD data into a low-dimensional space where clustering becomes feasible. Since k-means relies on Euclidean distance, rapidly interconverting conformations must map to nearby points in the reduced space; failing this leads to poor markovianity [12].

Feature selection. Two feature sets are used: contact distances (`hp35.mindists2`) and backbone dihedrals (`hp35.dihs.shifted`). Following [8], a Gaussian low-pass filter ($\sigma = 2\,\text{ns}$) is applied to each feature to suppress high-frequency recrossings between states.

Dimensionality reduction. Two linear methods are compared: PCA, which finds directions of maximal variance, and tICA, which finds directions of maximal autocorrelation at a lag time τ_{tICA}. PCA therefore captures the largest structural fluctuations, whereas tICA emphasizes the slowest dynamical processes: configurations far apart in tICA space tend to interconvert slowly, giving a closer correspondence between geometric and kinetic separation (see Supp. Methods B S1). Because MSMs aim to resolve slow kinetics, tICA is generally preferred. Combining the two feature sets with the two reduction methods gives four pipelines: *tica-distances*, *tica-dihedrals*, *pca-distances*, and *pca-dihedrals*. Since HP35 folds on the microsecond timescale, τ_{tICA} was inspected over 100–1000 frames ($\simeq$ 20–200 ns), well below the folding time, and fixed at 20 ns after checking that the leading tICA components were stable across this range. The number of retained components was chosen with an heuristic criterion so that the reduced coordinates can be expected to separate metastable states: relevant coordinates should show multimodal free-energy profiles and time series with long-lived plateaus separated by rapid jumps (Supp. Fig. SF1). This yielded 7 components for tica-dihedrals and 5 for the other three pipelines.

Clustering and lag-time selection. The reduced coordinates were clustered with k-means (k-means++ initialization, up to 100 iterations), testing $k = 30$, 50, and 100. The number of clusters and the lag time were chosen together, since they jointly control resolution and statistical reliability: more clusters provide higher resolution and speed up implied-timescale convergence, but leave fewer transitions per microstate and can disconnect the microstate network. The lag time must also stay well below the microsecond folding time [2], or intermediate kinetic details are lost. Because the implied-timescale test alone does not fix the lag time, the final choice was a compromise between timescale convergence, network connectivity, and adequate sampling of transitions, and so remains partly qualitative. Transition matrices were then estimated by maximum likelihood with a sliding-window scheme [7, Ch. 4.1] enforcing detailed balance, retaining the largest connected component as the state space. From a grid search with $n_{clusters} = \{30, 50, 100\}$ clusters and lag times between 10 and 100 ns; $n_{clusters} = 50$ and a lag time of 50 ns is selected for all four pipelines, as this seemed the best compromise between resolution, markovianity and connectivity (Supp. Fig. SF2 and SF3).

Spectral analysis and coarse-graining. The number of macrostates is chosen from the implied-timescale spectrum, taking the largest gap between successive timescales as the number of resolved slow processes. Microstates are then lumped into macrostates with PCCA++ (robust Perron-cluster cluster analysis) [7, Ch. 2.5.1–2.5.2]. The algorithm uses the sign pattern of the dominant eigenvectors of the transition matrix to identify the slow probability-exchange modes that govern relaxation to equilibrium, and groups microstates accordingly (see Supp. Methods B). The resulting macrostates are expected to correspond to the distinct metastable states of the system [7]. PCCA++ returns for each microstate i a vector of *fuzzy memberships* $\chi_{ik} \in [0, 1]$, the probability that i belongs to macrostate k (with $\sum_k \chi_{ik} = 1$); each frame inherits the memberships of its microstate. How assignments are used depends on the analysis and is stated in each case.

Dynamical coring. Assigning each frame to its highest-

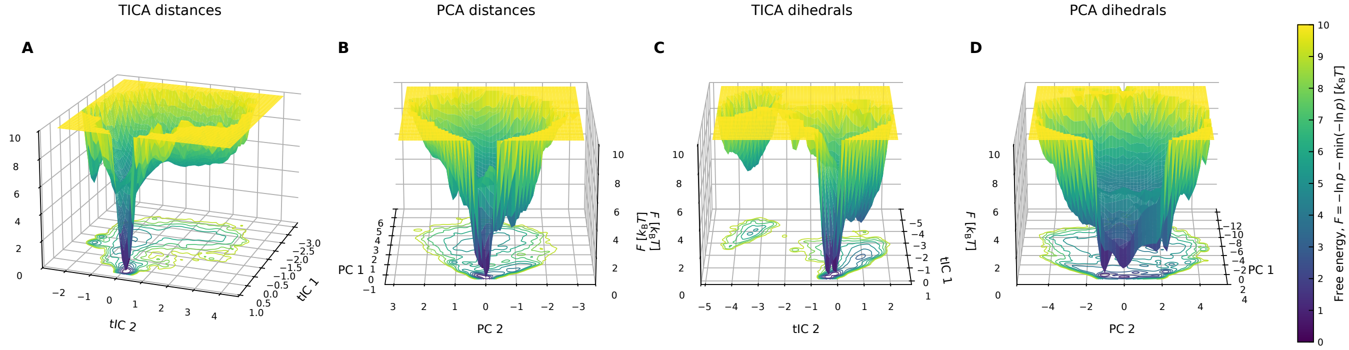

FIG. 2: ***Two-dimensional free-energy surfaces projected onto the first two reduced coordinates.*** *Because projection do not have a clear interpretation in terms of dynamics, these landscapes should be interpreted qualitatively. Nevertheless, substantial differences are observed among the four pipelines which later reflect differences in the metastable structure resolved by the MSMs. Distance-based models exhibit a single dominant minimum, whereas dihedral-based models show two major minima. The tICA projections additionally contain a low-free-energy basin well separated from the main funnel.*

membership macrostate produces a hard state trajectory, but near a state boundary rapid recrossings inflate the apparent number of transitions: these are more likely to reflect boundary noise than genuine interstate crossings. Dynamical coring [14] is a heuristic procedure to suppress them: starting from the hard-assigned trajectory, a transition into a new macrostate is registered only once the trajectory remains there for at least a threshold time $t_{\min}$, so that only long-lived occupancies survive. In this work I applied coring to the discretized macrostate trajectories with $t_{\min}$ equal to one lag time ($t_{\min} = \tau_{\mathrm{MSM}} = 50\,\mathrm{ns}$).

C. Statistical analysis

Chapman–Kolmogorov test. Model self-consistency was assessed with the CK test as implemented in Deeptime [10]. The microstate MSM at $\tau_{MSM} = 50\,\mathrm{ns}$ was coarse-grained into four metastable sets with PCCA++; for a range of lag times $k\tau_{MSM}$, the coarse-grained transition probabilities predicted by the reference model (propagated to $k\tau_{MSM}$) were compared with those estimated independently at each lag. Predicted and independent curves track closely and relax to consistent stationary values (Supp. Fig. SF4, shown for pca-dihedrals; the others behave similarly).

State populations. Macrostate populations are estimated in two ways. From the trajectory, after dynamical coring (above), as the fraction of frames assigned to each state. From the coarse-grained MSM, using the soft memberships as weights on the microstate stationary distribution, $P(S_k) = \sum_i \pi_i\,\chi_{ik}$, where π_i is the stationary probability of microstate i. The membership weighting downweights boundary microstates that cannot be confidently assigned to a single set.

State lifetimes. Macrostate lifetimes are likewise estimated in two ways. From the cored trajectory, as the average residence time in each state. From the coarse-grained transition matrix T, as $\langle t_k \rangle = \tau_{MSM}/(1 - T_{kk})$, the mean residence time of a single-exponential process leaving state k.

III. RESULTS

A. The four pipelines yield valid Markov models that already differ in their projected landscapes and timescales

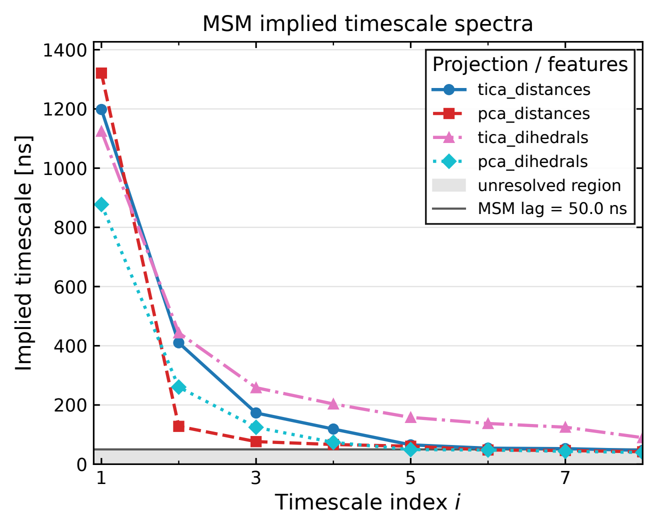

FIG. 3: ***Relaxation timescales spectra*** *at selected lag time $\tau_{MSM} = 50\,\mathrm{ns}$. The horizontal grey line marks the MSM resolution limit ($t = \tau_{MSM}$).*

Projecting each trajectory onto its first two reduced coordinates already shows that the pipelines describe the landscape differently (Fig. 2): the distance-based models yield a single dominant minimum, while both dihedral-based models resolve two, and the tICA projections show an additional well-separated basin. Very importantly, free energy surfaces do not generally have a simple interpretation in terms of dynamics (e.g. free energy maxima may not be kinetic barriers), therefore these surfaces are only indicative, but the differences already anticipate a pipeline dependence in the metastable structure resolved later. At the selected common discretization ($n_{\text{clusters}} = 50$, $\tau_{MSM} = 50\,\text{ns}$; see Methods), the implied-timescale test (Supp. Fig. SF2) shows that the slowest timescale is *approximately* converged for all four pipelines and falls in the microsecond range, consistent with the known folding time of HP35. Two observations: first, the implied timescales depend far more strongly on the pipeline (features + reduction method) than on the number of clusters. Second, the full relaxation spectra at $\tau_{MSM} = 50\,\text{ns}$ (Fig. 3) shows that the PCA pipelines yield systematically lower timescales than the tICA ones, indicating that they approximate the underlying slow dynamics less faithfully [13], as expected since PCA does not optimize for kinetic separation.

Taken together, these checks show that all four MSMs are approximately Markovian at $\tau_{MSM} = 50\,\text{ns}, n_{clusters} = 50$ and are expected to resolve the folding-unfolding transition as the slowest process, providing a valid common basis for the comparison that follows; at the same time, their timescale spectra already indicate that the pipelines disagree on how many slow processes exist beyond the dominant one.

B. The dominant process relates to the folding–unfolding transition

The dominant implied timescale is reasonably close across most pipelines (Fig. 3) and of the right order of magnitude (μs) to represent the folding–unfolding transition. A preliminary confirmation comes from the sign pattern of the first non-trivial right eigenvector, which closely follows the fraction of native contacts Q (Supp. Fig. SF5), suggesting that the slowest resolved mode separates folded from unfolded conformations.

The spectra differ, however, in what lies beyond that dominant process (Fig. 3). The PCA-distance model shows the largest gap between the first and second timescales, consistent with a strongly metastable, essentially two-state process. The other three show a less pronounced gap, and the tICA-dihedral model departs most from a two-state picture: its spectrum decays slowly, with several timescales remaining above the resolution limit ($t = \tau_{MSM}$) even though its slowest timescale matches the distance-based models.

To compare the models at the coarsest level, I impose a two-state PCCA++ decomposition on each and measure, for every model pair, the fraction of frames the two models place on opposite sides of the partition (the off-diagonal quadrants of Fig. SF6). At this level the four pipelines are *broadly* consistent: the two distance-based models disagree on only $\simeq 1.6\%$ of frames, and the distance/dihedral PCA pair on $\simeq 2.3\%$. The tICA-dihedral model is the most different, disagreeing with each of the others on $\simeq 8\%$ of frames.

For tICA-dihedral, the origin of this disagreement is visible in the single-model S_1 membership distributions (Supp. Fig. SF6, top): sharply bimodal and near $\{0, 1\}$ for the PCA-based representations, but broad and much fuzzier for tICA-dihedral. This fuzziness means the model does not resolve a clean boundary between the two basins—unsurprising, given that its spectrum lacks a clear eigenvalue gap. But the boundary sharpness is not the only issue: even the most agreeing model pairs retain a visible amount of probability mass in the off-diagonal corners (Supp. Fig. SF6, top)—frames that *both* models assign confidently, yet to opposite sides of the partition. This residual mass indicates that the four pipelines do not resolve exactly the same partition, beyond mere differences in boundary sharpness.

In summary, the pipelines agree on the coarsest description but not on its precise boundary, which was somewhat expected.

C. The folded basin is mostly defined by the hydrophobic core

The populations of S_0 and S_1 (Supp. Fig. SF6) closely match the unfolded and folded populations reported by Piana *et al.* [9], which is consistent with S_1 being the folded basin. To characterize it structurally while avoiding contamination from boundary frames, I define *consensus* microstates as those assigned to S_1 with high confidence ($\chi_{i,1} > 0.8$) by all four pipelines, and restrict the analysis to this subset ($\simeq 61\%$ of the trajectory).

A simple characterization follows from comparing, contact by contact, the distance distributions inside and outside the consensus set. For each contact present in the crystal structure (`hp35.crystaldists`), I score how strongly it distinguishes the two sets by the standardized median difference

$$\text{score}_{i,j} = \frac{\text{median}(r_{ij}^{\text{other}}) - \text{median}(r_{ij}^{\text{consensus}})}{\left[IQR_{ij}^{\text{other}} + IQR_{ij}^{\text{consensus}}\right]/2}. \quad (1)$$

A large positive score indicates a contact that is markedly shorter in the consensus set—relative to the spread of its distribution—and therefore characteristic of the folded basin.

The highest-scoring contacts (Fig. 4, bottom) are the long-range pair $(10, 28)$–$(10, 29)$, which packs helix 1 against helix 3, and the contacts among Phe6, Phe10 and Phe17 that form the hydrophobic core (Fig. 1) and residues close by. Intra-helix contacts, by contrast, do not rank among the discriminating features. This indicates

Model	N_{visits}	Mean (μs)	Median (μs)	$P(S_0)$	$P(S_1)$	ΔG_{1-0} (kcal mol^{-1})
tICA distances	35	2.712	1.725	0.311	0.689	-0.471
PCA distances	37	2.665	1.703	0.323	0.677	-0.438
tICA dihedrals	46	1.650	1.014	0.249	0.751	-0.655
PCA dihedrals	42	2.287	1.192	0.315	0.685	-0.461

TABLE I: *Empirical unfolded-state residence times computed from the cored macrostate trajectories, where N_{visits} is the number of unfolded visits observed in each trajectory. Relative free energies were calculated as $\Delta G_{1-0} = -k_{\mathrm{B}}T \ln[P(S_1)/P(S_0)]$ at $T = 360$ K.*

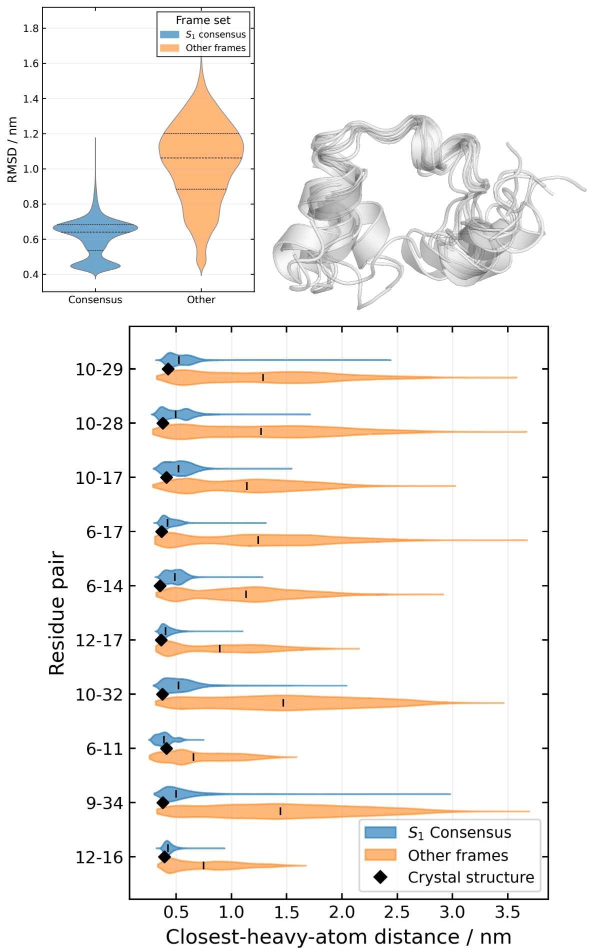

FIG. 4: **Structural analysis of S_1 consensus frames.** ***Top left***: *distribution of RMSD from the crystal structure (PDB 2F4K) for consensus frames (blue) and all other frames (orange); dashed lines mark the median and quartiles.* ***Top right***: *overlay of eight randomly chosen consensus frames made with Pymol [16].* ***Bottom***: *closest-heavy-atom distance distributions for the ten contacts that best discriminate the two frame sets, ranked by the score of Eq. 1 (most discriminating at the top). For each pair, the vertical tick marks the median of each distribution, and the black diamond marks the distance in the crystal structure.*

that appreciable helical structure persists in the unfolded ensemble, whereas hydrophobic-core formation is largely confined to the folded basin. The interpretation is corroborated by comparing all intra-helical contacts inside and outside the consensus set (Supp. Fig. SF7): outside, their distributions are much broader with systematically larger medians, yet still compatible with helical structure—especially for helix 3.

Interestingly, within the consensus ensemble both the RMSD and the $(10, 28)$–$(10, 29)$ contacts are bimodal (Fig. 4), which suggest the possibility that the folded basin contains two alternative conformations that differ in the relative positioning of the N- and C-termini. This is investigated in Sec. **??**.

Taken together, these results indicate that the folded and unfolded ensembles are distinguished primarily by the hydrophobic core and the helix-1–helix-3 packing that encloses it, rather than by helix formation *per se*.

D. Two-state thermodynamics are robust across pipelines but folding times are systematically underestimated

Interpreting S_0 and S_1 as the unfolded and folded basins, respectively, allows a rough estimate of the folding kinetics to be obtained. Although the resulting times should be regarded as indicative only, they provide a useful basis for comparing the different models. A more rigorous analysis would require computation of mean first-passage times and reactive fluxes on the full microstate MSM using transition path theory, which lies beyond the scope of the present work. To estimate lifetimes and populations from the trajectory, first I apply dynamical coring as explained in Methods. The resulting cored trajectory is shown in Supp. Fig. SF8. The folding time, t_{fold}, is then approximated as the average waiting time between successive unfolded-to-folded transitions in this cored trajectory. Results of folding times and relative populations are reported in Table I. The results are model dependent. Overall, trajectory-based estimates of the ΔG agree within one standard deviation with the original estimate of $-0.6 \pm 0.2\,kcal/\,mol^{-1}$ reported by *Piana et al.* [9] from direct analysis of the trajectory, while all models predict a shorter folding time (as the mean/median residence time in the state S_0) when compared to the value reported by [9] ($3.2 \pm 0.6\,\mu$s). Note that in principle another estimate can be derived directly from the PCCA coarse-grained transition matrix and fuzzy membership probabilities (see Suppl. Table II).

The thermodynamics are therefore robust across pipelines, whereas the kinetics are underestimated at this lag, most plausibly because the slowest timescale is not yet fully converged.

E. Three-states PCCA++ on tICA models reveal a misfolded state and a loop rearrangement

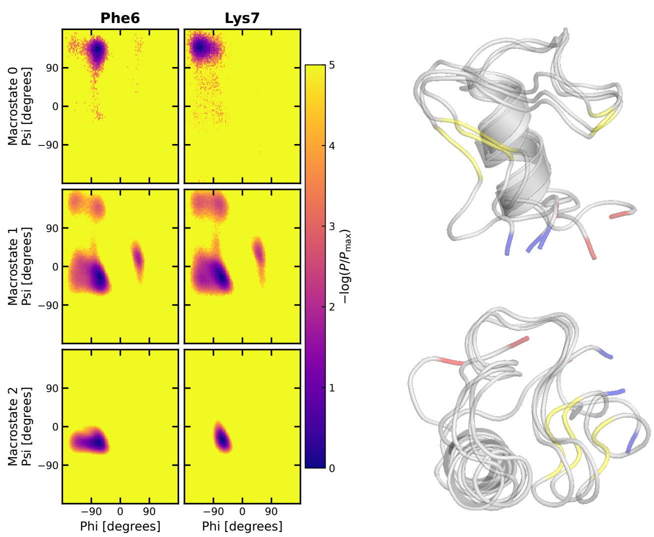

FIG. 5: ***Structural characterization of macrostate S_0 in the three-state tICA-distance model.*** *Ramachandran distributions of residues Phe6 and Lys7 together with representative conformations from macrostate S_0. This state is characterized by the formation of helix 3, while the N-terminal region adopts a β-hairpin-like structure in place of helix 1. Residues highlighted in yellow correspond to those shown in the Ramachandran plots. The N-terminus is highlighted in blue, the C-terminus in red. Visualization with Pymol [16].*

The implied timescale spectra of the tICA-based models (Fig. 3) suggest the presence of additional slow dynamical processes whose characteristic timescales are comparable to that of the dominant process. For both tICA models, the largest spectral gap occurs between the second and third implied timescales rather than between the first and second. This observation indicates that a three-state kinetic decomposition may provide a more natural description of the underlying dynamics than a simple two-state partition. Consequently, a three-state PCCA++ decomposition was applied to the tICA models, and the structural characteristics of the additional metastable state were investigated.

tICA-distances: A three-state PCCA partitioning produces a lowly populated ($\sim 1\%$) additional macrostate S_0. Key structural descriptors that distinguish this macrostate from the others are the dihedral angles of residues $Phe6, Lys7, Gly15$ which populate the β-region of the Ramachandran plot in S_0, while they populate both the α and the β region in S_1 and only the α_R region in S_2 (Fig.5). A visual inspection shows this is a well defined state, caracterized by helix 3 which never unfolds, compact structure and a stable beta-hairpin formed in the region of residues $6-14$. At all effects this appears to be a kinetic trap in the folding process. Inspection of the dynamically cored macrostate trajectory shows that this state is visited only once, where it remains occupied for approximately $1\,\mu$s.

tICA-dihedrals: A clear structural interpretation of macrostate S_0 is not immediately apparent. Comparison of the dihedral-angle distributions (Supp. Fig. SF9) indicates that the most pronounced difference between S_0 and the other two macrostates is localized in the $Leu20$–$Pro21$ loop connecting helices 2 and 3 in the native structure. In S_0, the ϕ/ψ angles of $Leu20$ predominantly occupy the β region of the Ramachandran plot, whereas in S_1 ($\simeq$ the unfolded basin) and S_2 ($\simeq$ the folded basin) they are concentrated in the α_R region. The transition $S_0 \leftrightarrow S_1$ therefore appears to correspond to a local conformational rearrangement of the second loop within the unfolded ensemble. Inspection of the dynamically cored macrostate trajectory shows that S_0 is visited multiple times, always from S_1. The state has a stationary population of approximately 3% and a median residence time of $0.32\,\mu$s.

F. Dihedral-based models resolve substructure within the folded basin

Interestingly, several observables, including the RMSD and specific long-range contacts such as 10–29, display bimodal distributions within the consensus folded ensemble (Section III C). The same bimodality is visible in the free-energy projections obtained from the dihedral-based models and is reflected in the distribution of the PCCA++ membership probability for macrostate 1. These observations suggest that the folded basin identified by the dihedral-based MSMs is itself heterogeneous and may contain two structurally distinct substates. But PCCA, at least with 3 states, did not identify finer structure within the folded basin. PCCA++ is a single algorithm , not the ground truth, so I will now move past its indications.

To investigate the possibility that the folded basin really harbours two distinct substates, I "manually" partition the folded ensemble identified by the dihedral-based models using the peaks in the PCCA++ membership probabilities (see Supp. Figure SF6) and compare their structural characteristics. The two sets of microstates

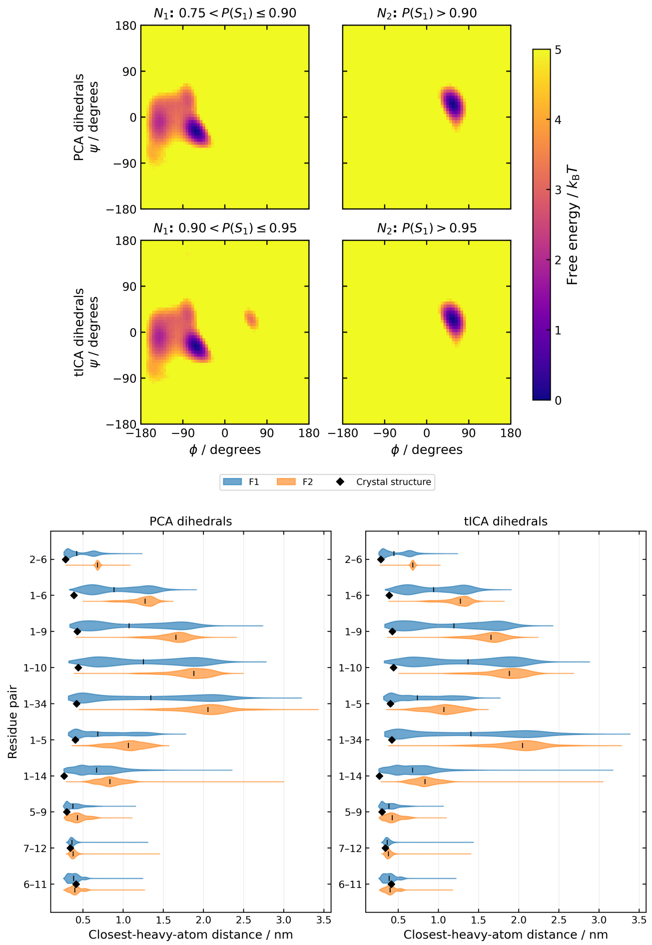

FIG. 6: ***Structural distinction between the two folded substates identified by the dihedral-based MSMs:*** *(ϕ, ψ) distribution of the residue Asp3 in the two subpopulations corresponding to the distinct minima observed within the folded basin.*

are found to be distinguished primarily by the conformational state of a single residue, *Asp3*. In one state, *Asp3* predominantly occupies the α_R region of Ramachandran space, whereas in the other it preferentially populates the α_L region (Fig. 6). From a more detailed analysis, it turns out that these substates actually do qualify as metastable, with a dwell time of $\simeq 0.3\mu s$ (Supp. Table: III) which is totally comparable with the dwell time of the third macrostate identified by pcca on tica-dihedrals and, even more notably, a very nice single-exponential distirbution (Supp. Fig: SF10). So even if PCCA did not find it, this partition is important and no less important than the one that it found. Concluding the analysis on this results, it demostrates the limitations of making partitioning the state space relying on a single algorithm (PCCA in this case). This analysis suggests that the folded basin is is itself heterogeneous, split by the Asp3 backbone conformation which however do not appear among the slowest processes. So this analysis, differently from the previous ones, is not an analysis of macrostates. PCCA, at least with $n \leq 3$ states, does not identify the interconversion between these two substates as one of the slow processes. I want to check the lifetimes of these substates, to see if they qualify as "metastable" even if PCCA hasnt found them. At the end, PCCA is an algorithm, it is not the ground truth.

IV. DISCUSSION AND CONCLUSIONS

This project set out to characterize the conformational ensemble of the HP35 Nle/Nle mutant from a structural standpoint. To limit the bias inherent in any single MSM pipeline, four models were built and compared, varying the feature set and the dimensionality-reduction step.

Taken together, the four models yield the following picture of the HP35 folding landscape:

1. A dominant free-energy barrier separates two metastable basins: an unfolded basin ($\simeq$ 31–32%) and a folded/partially folded basin ($\simeq$ 67–68%). The two are distinguished not by secondary-structure content but mainly by the presence or absence of the native hydrophobic-core contacts and the close packing of helices 1 and 3 (Sec. III C, III D);

2. A lowly populated ($\simeq$ 1%), compact metastable state with a lifetime of $\sim 1\,\mu$s, in which helix 3 and the helix-2–helix-3 loop are folded but an extended, hairpin-like segment replaces helix 1 (Sec. III E);

3. Two substates within the folded basin, F_1 and F_2, of comparable population (33.8–35% and 31–32%, respectively). They differ in the configuration of the N-terminus and are discriminated mainly by the (ϕ, ψ) angles of Asp3 (α_R in F_1, α_L in F_2) and by the presence (F_1) or absence (F_2) of several native contacts involving Leu1 (Sec. III F).

This picture agrees on several points with the macrostate model of Jain and Stock [4], obtained from a MSM construction pipeline which used dihedral representation and with most-probable-path clustering. Their intermediate I_1 corresponds closely to the substate F_2 found here. They also report an off-pathway, partially folded state I_3 in which an extended conformation replaces helix 1; this matches the putative kinetic trap resolved by my tICA–distances model, both structurally and in population (1.49% vs. $\simeq$ 1% here). On these points the present analysis supports their description, and it further shows that I_3 is recoverable with a standard tICA/PCCA++ pipeline—here, specifically when the features are contacts rather than dihedrals.

On the methodological side, this project is a concrete illustration of how strongly the choice of MSM pipeline can shape the results, even for a small protein and a high-quality dataset. Apart from the dominant two-state transition, on which all four pipelines agree, the finer structure they resolve differs markedly: the kinetic trap (Sec. III E) was recovered only by the tICA–distances model, and the two folded-basin substates only by the two dihedral-based models (Sec. III F). This is not suprising, as any single pipeline reduces the dynamics to one particular low-dimensional representation, the resulting models are therefore best read not as competing definitive descriptions but as complementary views of the same underlying dynamics.

A related point is the limitation of relying on PCCA++ alone for coarse-graining, especially when—as here—the implied-timescale spectrum shows no clear gap. PCCA++ failed to separate the two folded-basin substates even though their dwell time is comparable to that of the loop rearrangement it did resolve. This highlights that metastability (a long dwell time) and spectral slowness (what PCCA++ ranks) are related but not identical: a state can be long-lived without its interconversion being among the slowest processes, and such states are then easily missed by a purely spectral coarse-graining [6].

Several possibilities remain for extending this analysis. The single most useful next step would be to test how robust the results are to the metastable-state identification procedure itself—for instance by comparing PCCA++ with alternatives such as MPP which was used in several MSM studies of HP35 [4, 6, 8]. A more complete kinetic characterization would, in turn, require constructing the full microstate MSM and applying transition-path-theory (TPT) analysis.

V. REFERENCES

Data availability statement: Preprocessed trajectories of the HP35 villin headpiece are kindly made available by Nagel et al. and downloadable at their GitHub https://github.com/moldyn/HP35-DESRES. All code necessary to replicate this analysis is available on GitHub at https://github.com/miriamzara/MSM.

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Appendix A: Supplementary Figures

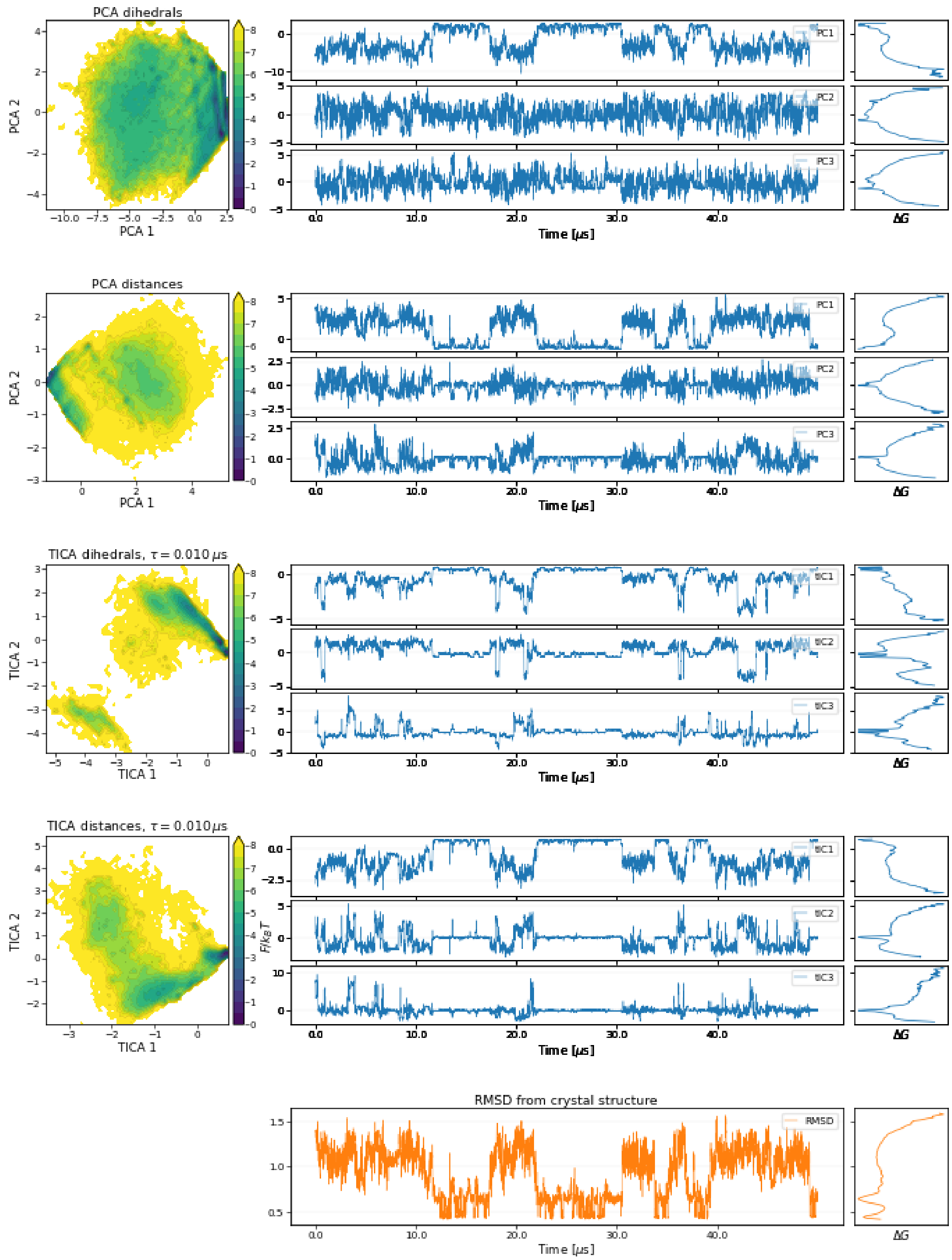

FIG. SF1: **Reduced-coordinate representations obtained from the four feature–dimensionality-reduction pipelines.** Rows correspond to PCA dihedrals, PCA contact distances, tICA dihedrals, and tICA contact distances. For each pipeline, the left panel shows the projection of the trajectory onto the first two reduced coordinates. The remaining panels show the time evolution of the first three reduced coordinates during the first $\sim 40\,\mu$s of simulation together with the corresponding one-dimensional free-energy profiles computed from the full trajectory. The bottom row reports the RMSD from the crystal structure and its free-energy profile for comparison.

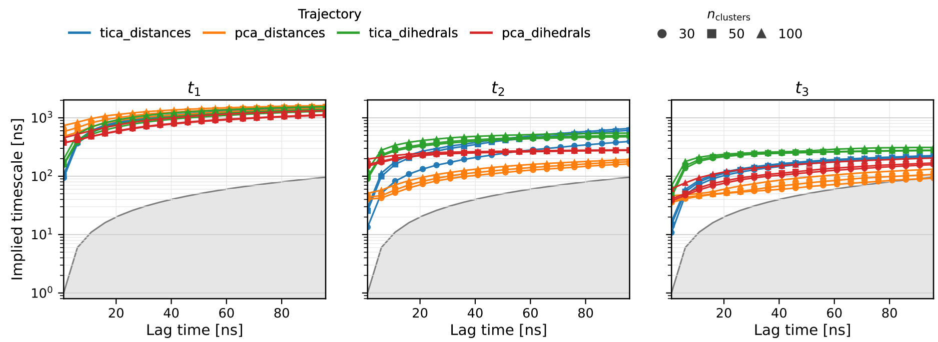

FIG. SF2: ***Results of the timescale convergence test**: First three implied timescales as a function of the MSM lagtime. Colors denote the four combinations of feature set and dimensionality reduction method, while marker styles indicate the number of microstates used in the state-space discretization. The shaded region corresponds to $t_i \leq \tau_{MSM}$, where t_i is the implied timescale and τ_{MSM} the MSM lagtime. Timescales falling within this region are not resolved by the model, as the associated relaxation process occurs on a timescale shorter than a single Markov-chain transition step.*

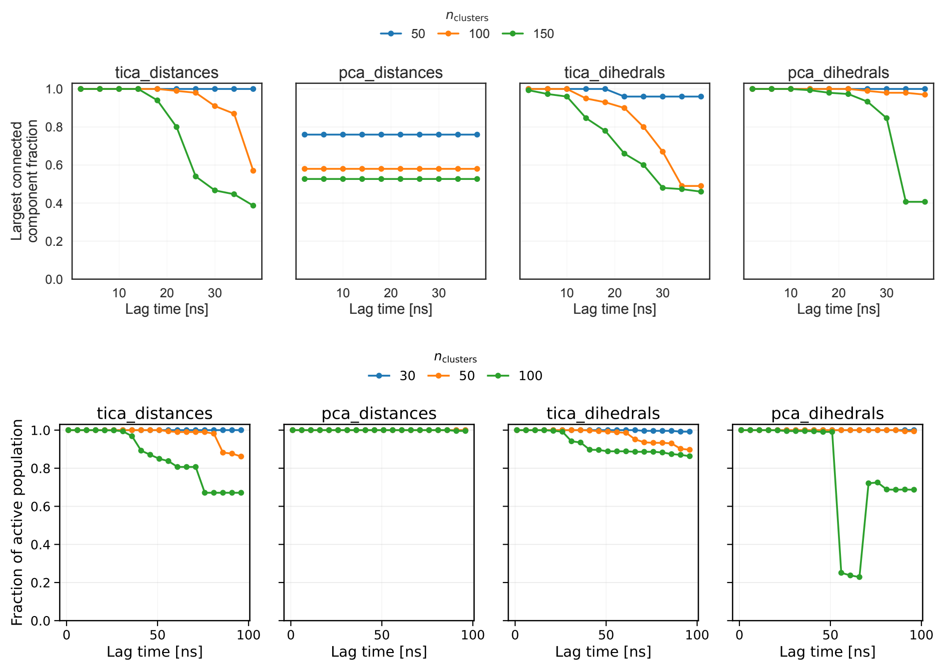

FIG. SF3: **MSM connectivity and active population as a function of lag time.** Top panel: fraction of microstates in the largest connected component of the MSM as a function of lag time. Bottom panel: fraction of the total equilibrium population contained in the largest connected component of the MSM as a function of lag time

Model	$P(S_0)$	$P(S_1)$	$\langle t_0 \rangle$ (μs)	$\langle t_1 \rangle$ (μs)	ΔG_{1-0} (kcal mol^{-1})
tICA distances	0.230	0.770	1.591	5.312	-0.863
PCA distances	0.303	0.697	1.933	4.446	-0.596
tICA dihedrals	0.218	0.782	1.471	5.286	-0.915
PCA dihedrals	0.358	0.642	1.407	2.520	-0.417

TABLE II: **Thermodynamic and kinetic properties estimated from two-state Markov Models:** Stationary populations, mean residence times, and relative free energies obtained from the two-state Markov model induced by the PCCA++ coarse graining of each MSM. Mean residence times were computed as $\langle t_i \rangle = \tau/(1 - T_{ii})$, where T denotes the coarse-grained transition matrix and τ the MSM lag time. Relative free energies were calculated as $\Delta G_{1-0} = -k_{\mathrm{B}}T \ln[P(S_1)/P(S_0)]$ at $T = 360$ K.

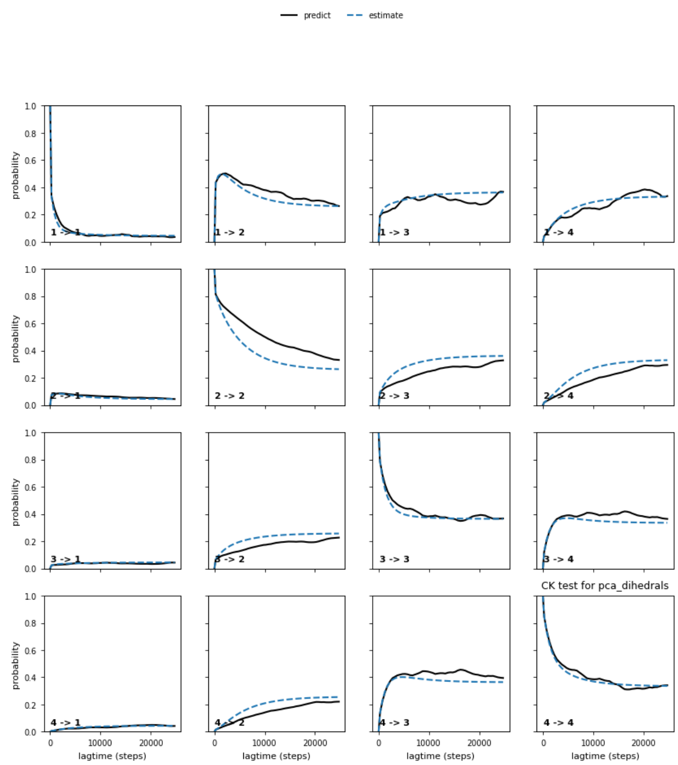

FIG. SF4: **Chapman–Kolmogorov test for the pca-dihedrals MSM**. The microstate model estimated at $\tau = 50\,\mathrm{ns}$ was lumped into four metastable sets with PCCA++. Each panel $(i \to j)$ shows the probability of being in set j at time $k\tau$ given the system started in set i. Solid lines: MSMs estimated independently at each lag; dashed lines: predictions from the $\tau = 50\,\mathrm{ns}$ model. Close agreement indicates approximate Markovianity at the chosen lag.

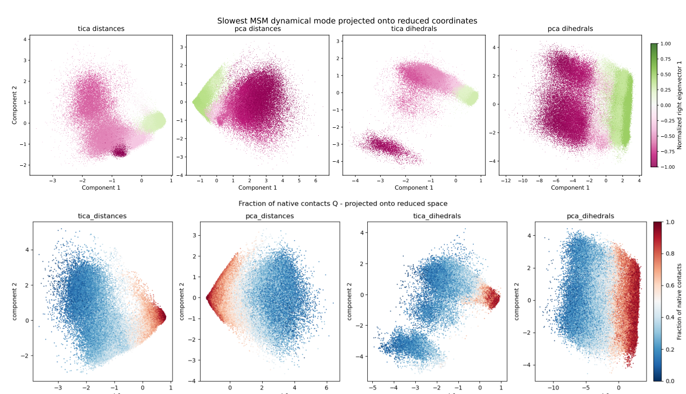

FIG. SF5: **Comparison between the slowest MSM relaxation mode and a structural folding descriptor.** Top row: values of the first non-trivial eigenvector projected onto the first two components of the reduced coordinate space. Positive and negative components identify regions that exchange probability on the slowest resolved timescale of the Markov model. Bottom row: projection of the fraction of native contacts, Q, onto the same coordinates. The close correspondence between the eigenvector sign pattern and the distribution of Q indicates that the slowest kinetic process captured by the MSM is associated with the folding-unfolding transition.

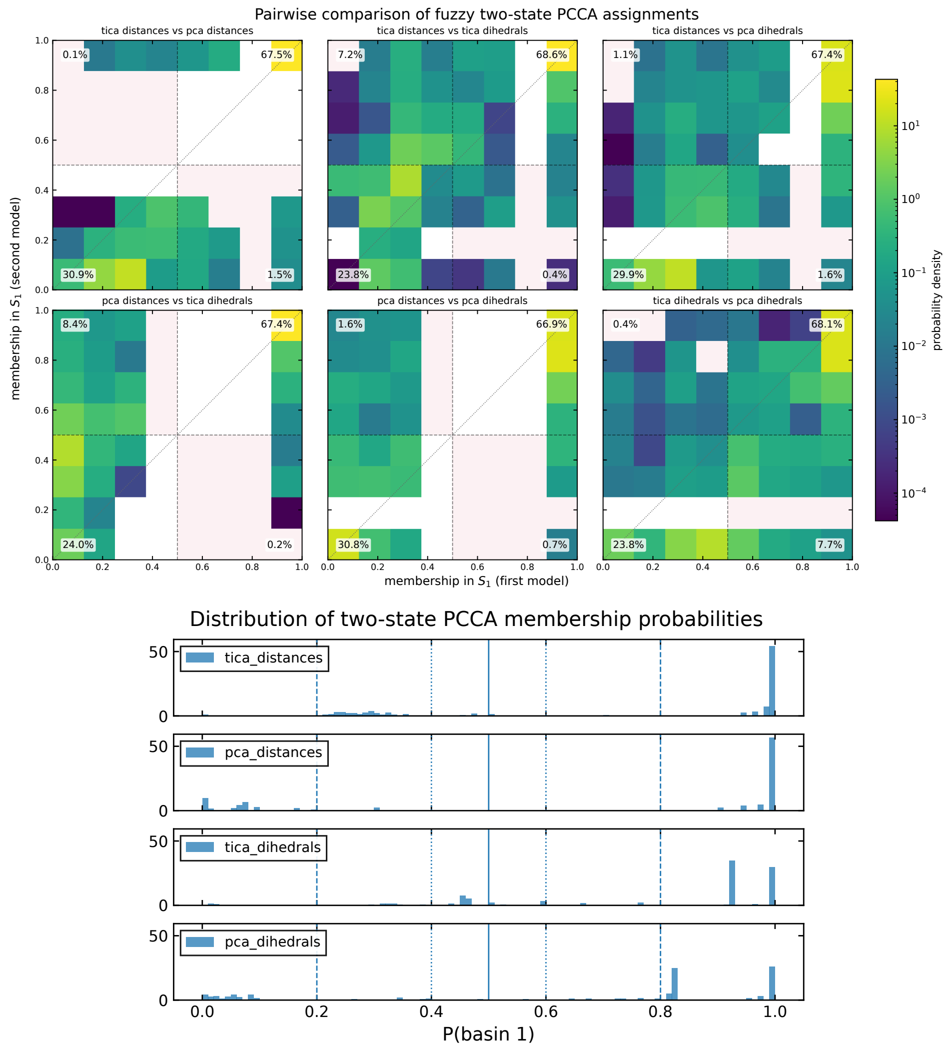

FIG. SF6: *Fuzzy membership probabilities of the trajectory frames to the two macrostates across models. Top: Each panel shows the fuzzy membership probability of each frame in the trajectory to macrostate S_0 ($\simeq$ denatured basin) and macrostate S_1 ($\simeq$folded basin) for a different model. The models based on distances (tica-distances and pca-distances) show very similar distributions, while the tica-dihedrals model exhibits a broader distribution, particularly for frames assigned to the denatured basin. This indicates that the tica-dihedrals model has more uncertainty in assigning frames to the denatured basin. Bottom: Distribution of the fuzzy membership probability of each frame to macrostate* 1 *for the four MSMs. Probabilities close to* 0 *or* 1 *correspond to frames that are confidently assigned to a single macrostate.*

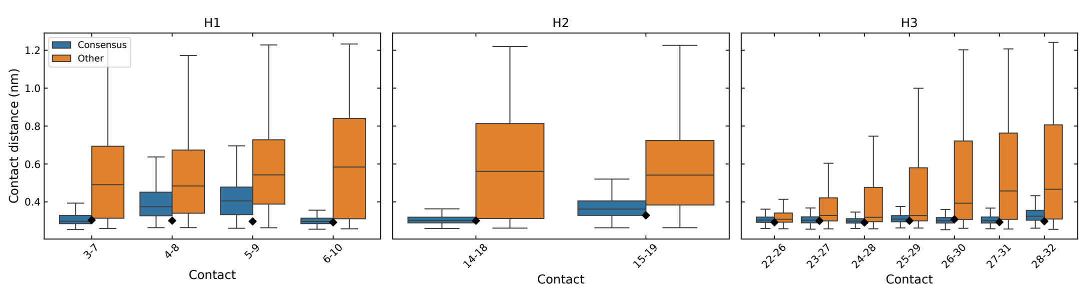

FIG. SF7: **Helix content in the consensus S_1 frames:** Distribution of distances for all intra-helical contacts in the consensus folded ensemble (S_1, blue) and in all other frames (orange). The black squares marks the distance observed in the crystal structure PDB 2F4K. Although helical contacts are generally more compact within S_1, their distributions overlap substantially with those of the remaining frames, indicating that significant helical structure is present both inside and outside the folded basin.

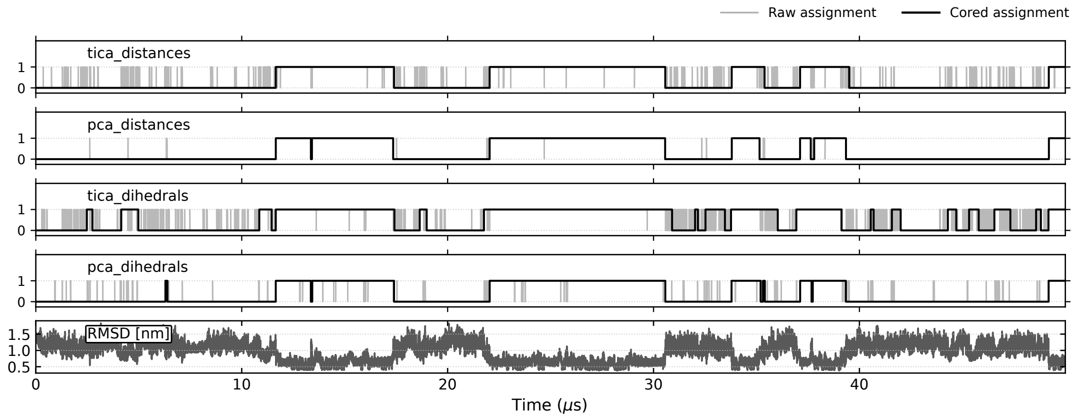

FIG. SF8: **Comparison of raw and filtered macrostate trajectories for the two-state PCCA++ models.** For each MSM, the grey trace shows the instantaneous hard assignment of each frame, while the black trace shows the filtered assignment obtained by removing transitions that persist for less than one Markov lag time ($\tau = 50\,\mathrm{ns}$). The RMSD time series is shown at the bottom for reference. This filtering suppresses rapid recrossings near the state boundary and highlights the long-lived folded and unfolded episodes underlying the dynamics. Only the first $\simeq 50\,\mu s$ of the trajectory are shown.

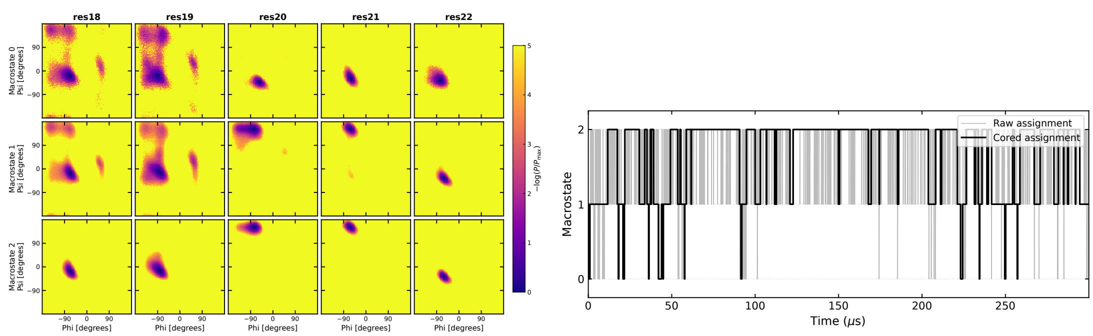

FIG. SF9: **Structural caracterization of the 3-state PCCA macrostates for the tica-dihedrals model:** Top: (ϕ, ψ) distribution of the residues $18 - 22$ in the three macrostates S_0, S_1, S_2. Bottom: trajectory of the three-state PCCA++ model for tICA-dihedrals, with hard assignations and cored assignations (transitions are counted if they remain in the destination for at least one markov step, which is $\tau = 50\,ns$).

Model	State	N_{visits}	Mean [μs]	Median [μs]	Population
tica_dihedrals	F1	177	0.603	0.463	0.350
tica_dihedrals	F2	145	0.658	0.435	0.313
pca_dihedrals	F1	195	0.529	0.412	0.338
pca_dihedrals	F2	145	0.676	0.480	0.321

TABLE III: Dwell times in the folded sub-basins F_1 and F_2, computed from the cored microstate trajectory of each dihedral model. N_{visits} is the number of continuous visits to the sub-basin, and the population is the fraction of cored frames assigned to it.

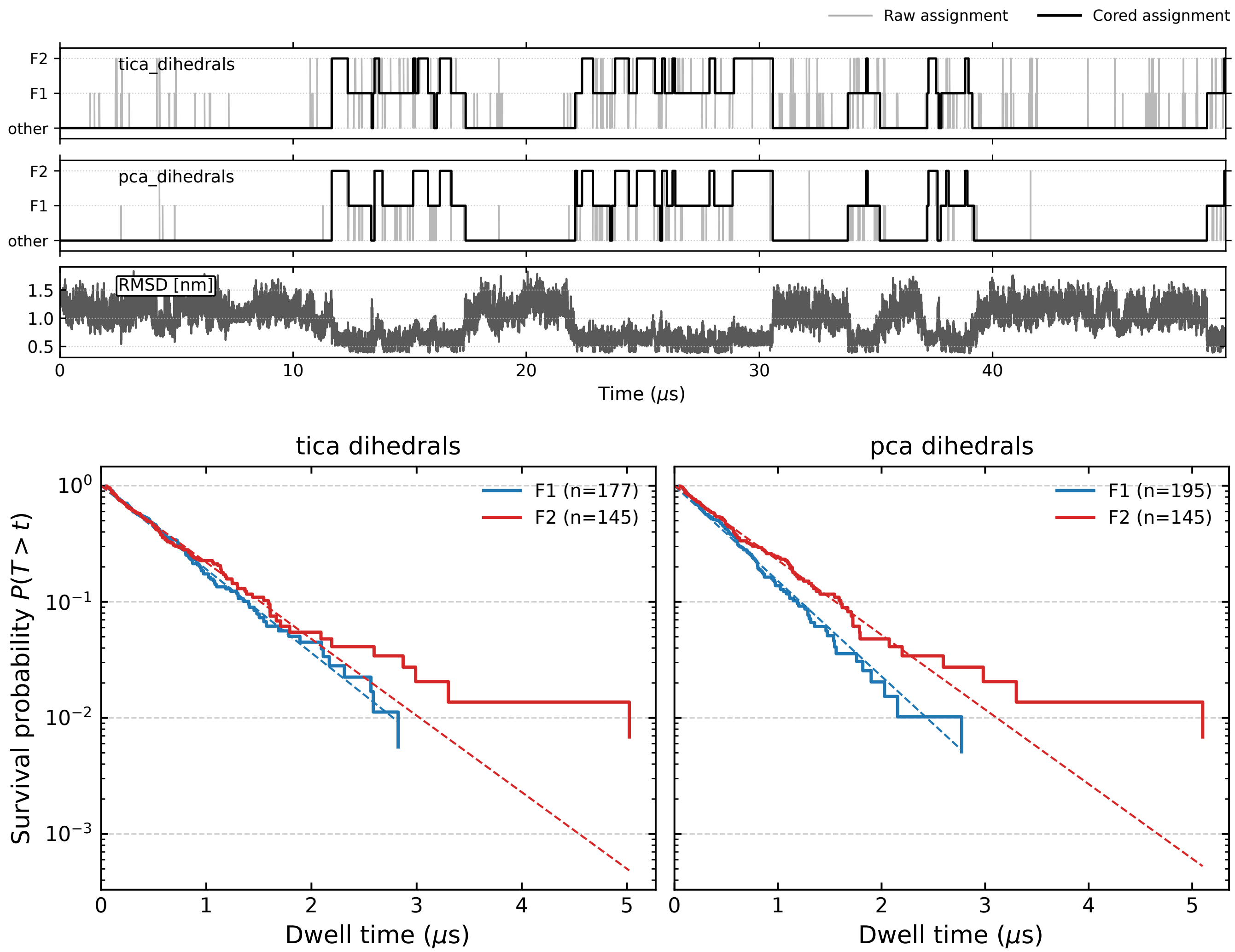

FIG. SF10: **Analysis of subsets F_1 and F_2 identified by dihedral-based models. Top**: first $\simeq 40\,\mu s$ of the discretized trajectory (F_1, F_2, other) highlight $F_1 \leftrightarrow F_2$ transitions. **Bottom**: survival probability (1-cdf) and linear fit shows the distribution of the residence time is well approximated by an exponential.

Appendix B: Supplementary Methods

Here I provide the technical details of the procedure summarized in Methods.

S1. Derivation of PCA and tICA

The presentation of tICA in this section follows closely that of [15].

PCA and tICA are both linear dimensionality reduction techniques, i.e. they project the data onto a linear subspace of the original feature space. Let $\{\mathbf{x}(t)\}_{t=0}^{T-1}$ be a multidimensional, discrete time-series, where each snapshot is a column vector, of dimension D, corresponding to an arbitrary, vectorized representation of a protein conformation (e.g. a set of dihedral angles). Suppose the data has been sampled from a *stationary* and *time-reversible* stochastic process. Without loss of generality, I assume that the data have been centered, $\mathbb{E}[\mathbf{x}] = 0$. Using the braket notation $\langle x| := \mathbf{x}^T$, $|x\rangle := \mathbf{x}$, I define the covariance matrix:

$$\mathbf{C}_0 := \mathbb{E}[|x\rangle \langle x|] \tag{B1}$$

and the time-lagged covariance matrix:

$$\mathbf{C}_{0,\tau} := \mathbb{E}\left[|x(t)\rangle \langle x(t+\tau)|\right]. \tag{B2}$$

For a stationary reversible process,

$$\mathbf{C}_{0,\tau} = \mathbf{C}_{\tau,0}^T = \mathbf{C}_{\tau,0},$$

so that the lagged covariance matrix may be denoted simply by $\mathbf{C}_\tau$.

If $|\hat{u}\rangle \langle \hat{u}|$ is a projection operator (i.e. $|\hat{u}\rangle$ is a unit vector), then:

$$\langle \hat{u}| \mathbf{C}_0 |\hat{u}\rangle = \mathbb{E}[\langle \hat{u}|x\rangle \langle x|\hat{u}\rangle] = \mathbb{E}[\langle \hat{u}|x\rangle^2] \tag{B3}$$

is the variance of the projection of the data along $|\hat{u}\rangle$, and

$$\begin{aligned}\mathrm{ACF}_{\hat{u}}(\tau) &= \frac{\langle \hat{u}| \mathbf{C}_\tau |\hat{u}\rangle}{\langle \hat{u}| \mathbf{C}_0 |\hat{u}\rangle} \\ &= \frac{\mathbb{E}[\langle \hat{u}|x(t)\rangle \langle \hat{u}|x(t+\tau)\rangle]}{\mathbb{E}[\langle \hat{u}|x\rangle^2]}\end{aligned} \tag{B4}$$

its autocorrelation at lag time τ.

PCA seeks the direction of maximal variance. The first principal component is defined as

$$\begin{aligned}|\hat{u}_1\rangle &= \arg\max \langle \hat{u}| \mathbf{C}_0 |\hat{u}\rangle \\ &\quad \text{subject to } \|\hat{u}\| = 1\end{aligned} \tag{B5}$$

this optimization problem can be solved introducing a Lagrange multiplier λ, and the stationarity condition yields

$$\mathbf{C}_0 |\hat{u}_i\rangle = \lambda_i |\hat{u}_i\rangle, \tag{B6}$$

i.e. the projection versor is an eigenvector of the covariance matrix. The corresponding eigenvalue equals the variance of the data projected onto the principal component:

$$\langle \hat{u}_i| \mathbf{C}_0 |\hat{u}_i\rangle = \lambda_i \cdot \langle \hat{u}_i|\hat{u}_i\rangle = \lambda_i.$$

Then the second principal component is defined as the direction of maximal variance orthogonal to the first, giving the optimization problem:

$$
\begin{aligned}
|\hat{u}_2\rangle = &\text{argmax} \langle u| \mathbf{C}_0 |u\rangle \\
&\text{subject to } \langle u|u\rangle = 1 \\
&\text{and } \langle u|\hat{u}_1\rangle = 0
\end{aligned}
\tag{B7}
$$

which again can be solved with Lagrange multipliers. Iterating, this implies that the principal components form an orthonormal basis of eigenvectors of the covariance matrix, i.e. they diagonalize the covariance matrix. Dimensionality reduction ($d < D$) is achieved by taking the components along the first d principal components $\{|\hat{u}_1\rangle, \ldots, |\hat{u}_d\rangle\}$.

$$
z_i = \langle \hat{u}_i|x\rangle, \qquad \mathbf{z} = U^T\mathbf{x}. \tag{B8}
$$

This is a linear projection onto an ortonormal basis:

$$
|x\rangle \to |z\rangle := \sum_{i=1}^{d} \langle x|\hat{u}_i\rangle \, |\hat{u}_i\rangle
$$

By construction, in the reduced space components are uncorrelated and ordered by decreasing variance:

$$
\mathbb{E}[|z\rangle \langle z|] = \text{diag}(\lambda_1, \cdots \lambda_d) \tag{B9}
$$

tICA Unlike PCA, tICA [15] seeks directions $|u\rangle$ that maximize the autocorrelation Eq.(B4) at a fixed lag time τ. The lagtime τ is a free parameter. The autocorrelation is invariant under scaling of the projection vector $|u\rangle$. Differently from PCA, the normalization constraint is usually implemented as $\langle u| \mathbf{C}_0 |u\rangle = 1$, as this choice simplifies the optimization problem. Formally, tICA solves the optimization problem:

$$
\begin{aligned}
|\hat{u}_1\rangle = &\text{argmax} \langle u| \mathbf{C}_\tau |u\rangle \\
&\text{subject to } \langle u| \mathbf{C}_0 |u\rangle = 1
\end{aligned}
\tag{B10}
$$

using the same procedure as for PCA, I can solve this with Lagrange multipliers, and show that

$$
\mathbf{C}_\tau |\hat{u}_1\rangle = \lambda_1 \mathbf{C}_0 |\hat{u}_1\rangle, \tag{B11}
$$

.

The generalized eigenvalue λ_i is equal to the autocorrelation of the projected coordinate $\langle \hat{u}_i|x(t)\rangle$ at lag time τ. Assuming that this coordinate approximates a single relaxation mode of the dynamics, its autocorrelation decays exponentially, yielding

$$
\lambda_i \approx e^{-\tau/t_i}, \tag{B12}
$$

where t_i is the corresponding relaxation timescale. In practice, the relation is approximate because a tIC may contain contributions from multiple dynamical modes.

Similarly as done with PCA, I can iterate this procedure and define the second tIC as the direction of maximal autocorrelation at lag time τ that is *uncorrelated* with the first:

$$
\begin{aligned}
|\hat{u}_2\rangle = &\text{argmax} \langle u| \mathbf{C}_\tau |u\rangle \\
&\text{subject to } \langle u| \mathbf{C}_0 |u\rangle = 1 \\
&\text{and } \langle u| \mathbf{C}_0 |\hat{u}_1\rangle = 0
\end{aligned}
\tag{B13}
$$

the solution $|\hat{u}_1\rangle$ is again a solution of the same generalized eigenvalue problem as in Eq.(B10). generalized eigenvalue problem. Dimensionality reduction ($d < D$) is achieved by taking the components of the data $\mathbf{x}(t)$ along the first d tICs $\{|\hat{u}_1\rangle, \ldots, |\hat{u}_d\rangle\}$.

$$\mathbf{z}(t) = (\langle x(t)|\hat{u}_i\rangle, \cdots, \langle x(t)|\hat{u}_d\rangle) \tag{B14}$$

By construction, in the reduced space components z_i have unit variance, are uncorrelated at zero lag, and are ordered by decreasing autocorrelation at lag time τ.

$$\mathbb{E}[|z\rangle\langle z|] = \mathbb{I} \tag{B15}$$

tICA as a diagonalization in the whitened coordinate space It is worth noting that unlike in PCA, the tIC vectors are generally not orthogonal under the standard Euclidean inner product. Instead, as I said, they satisfy the generalized orthogonality relation

$$\langle u_i| \mathbf{C}_0 |u_j\rangle = \delta_{ij}.$$

i.e. they corresponding components z_i and z_j are *uncorrelated* at zero lag. Nevertheless, the tICA procedure can be reformulated as a standard eigenvalue problem in a transformed coordinate system. The key idea is to first apply a whitening transformation, which rescales and rotates the original data so that all directions have unit variance and are mutually uncorrelated. Define the ZCA whitening transformation

$$|y\rangle = \mathbf{C}_0^{-1/2} |x\rangle$$

where $C_0^{-1/2}$ is symmetric and positive definite. By construction, the covariance matrix in the transformed space is the identity:

$$\mathbb{E}[|y\rangle\langle y|] = \mathbf{C}_0^{-1/2}\mathbf{C}_0\mathbf{C}_0^{-1/2} = \mathbf{I}.$$

and time-lagged covariance matrix in the whitened space is

$$\hat{\mathbf{C}}_\tau = \mathbf{C}_0^{-1/2}\mathbf{C}_\tau\mathbf{C}_0^{-1/2}$$

$\hat{\mathbf{C}}_\tau$ is symmetric and can thus be diagonalized:,

$$\hat{\mathbf{C}}_\tau |v_i\rangle = \lambda_i |v_i\rangle, \qquad \langle v_i|v_j\rangle = \delta_{ij}.$$

If $|u_i\rangle$ solves the tICA problem (B10), then $|v_i\rangle = \mathbf{C}_0^{1/2} |u_i\rangle$ is an eigenvector of $\hat{\mathbf{C}}_\tau$ with the same eigenvalue.
Indeed, multiplying the generalized eigenvalue problem (B10) from the left by $\mathbf{C}_0^{-1/2}$ and defining $|v_i\rangle = \mathbf{C}_0^{1/2} |u_i\rangle$ yields

$$\hat{\mathbf{C}}_\tau |v_i\rangle = \lambda_i |v_i\rangle.$$

Consequently, the tICA components z_i correspond to the ortogonal projection of the whitened data $\mathbf{y}$ onto the ortonormal basis defined by $\{|v_i\rangle\}$:

$$z_i(t) = \langle x(t)|u_i\rangle = \langle x(t)|C_0^{-1/2}v_i\rangle = \langle y(t)|v_i\rangle.$$

Alternative generalized eigenvector normalizations for tICA
The construction presented above corresponds to the original formulation of tICA, in which the covariance matrix of

the transformed data is the identity (B15). HoIver, alternative rescaling of the transformed coordinates have been used in the literature and are usually implemented in softwares to endow Euclidean distances in the reduced space with a more direct kinetic interpretation. These can be particularly useful when the tICA coordinates are subsequently clustered to construct a Markov state model, as I do in this work.

The most commonly used choice is the *kinetic map* in which each tIC component z_i is multiplied by its corresponding eigenvalue,

$$\tilde{z}_i(t) = \lambda_i z_i(t). \tag{B16}$$

With this normalization, the coordinates representing the sloIst processes "Iight more", and a unit step in those directions in the reduced space is assigned a larger distance. An alternative is the *commute map*, where each coordinate is rescaled according to its implied timescale,

$$\tilde{z}_i(t) = \sqrt{\frac{t_i}{2}} z_i(t), \qquad t_i = -\frac{\tau}{\ln \lambda_i}. \tag{B17}$$

With this normalization, Euclidean distances approximate commute distances, i.e. the expected time required to travel from one configuration to another and back. This is a variation on the same idea behind the kinetic map, being eigenvalues and timescales connected by (B12).

S2. Elements of Markov theory

Transition matrix and spectral decomposition. A Markov state model approximates the conformational dynamics as a memoryless jump process on the discrete microstates, encoded in the transition matrix $T(\tau)$ estimated at a fixed lag time τ. Following the convention used in DEEPTIME [**?**], T is taken to be row-stochastic,

$$\sum_j T_{ij} = 1, \qquad T_{ij} \geq 0, \tag{B18}$$

so that T_{ij} is the conditional probability of being in state j one lag step after being in state i. Probability distributions are represented as row vectors and evolve by right multiplication,

$$p^{(n)} = p^{(0)} T^n. \tag{B19}$$

The stationary distribution π is the left eigenvector with unit eigenvalue, $\pi T = \pi$ with $\sum_i \pi_i = 1$, and the estimator used here enforces detailed balance, $\pi_i T_{ij} = \pi_j T_{ji}$.

Assuming T is diagonalisable, it admits the spectral decomposition

$$T = \sum_{i=1}^{N} \lambda_i \, r_i \, l_i^{\mathsf{T}}, \tag{B20}$$

where the eigenvalues are ordered as $1 = \lambda_1 > \lambda_2 \geq \ldots$, and r_i, l_i are the right and left eigenvectors,

$$T r_i = \lambda_i r_i, \qquad l_i^{\mathsf{T}} T = \lambda_i l_i^{\mathsf{T}}, \tag{B21}$$

normalised to be bi-orthonormal, $l_i^{\mathsf{T}} r_j = \delta_{ij}$. The leading pair is fixed by $r_1 = \mathbf{1}$ (the constant vector, since T is row-stochastic) and $l_1 = \pi$. Substituting (B20) into (B19) and using bi-orthonormality, any initial distribution relaxes to equilibrium as a sum of exponentially decaying modes,

$$p^{(n)} = \sum_i c_i \, \lambda_i^n \, l_i^{\mathsf{T}}, \qquad c_i = p^{(0)} r_i. \tag{B22}$$

The dominant term ($\lambda_1 = 1$) is the stationary distribution π; because $|\lambda_i| < 1$ for $i \geq 2$, every other mode decays with a characteristic relaxation (implied) timescale

$$t_i = -\frac{\tau}{\ln \lambda_i}. \tag{B23}$$

Relaxation modes and probability flux. The left eigenvectors $\{l_i\}_{i\geq 2}$ describe the spatial structure of the relaxation processes: states whose components have opposite sign exchange probability on the associated timescale t_i, and the magnitude of each component measures how strongly that state participates in the process. The redistribution driven by these modes over a single lag step is obtained by expanding a distribution in the same basis, $p = \sum_i c_i l_i^\mathsf{T}$ with $c_i = p\, r_i$, and forming the difference

$$\Delta p = pT - p = \sum_{i\geq 2} c_i\,(\lambda_i - 1)\, l_i^\mathsf{T}, \tag{B24}$$

where the stationary term drops out because $\lambda_1 = 1$. Equation (B24) makes explicit that only the sub-unit eigenvalues carry probability flux, that the slowest modes ($\lambda_i \to 1$) produce the smallest per-step change and therefore dominate the long-time relaxation, and that the geometric pattern of each flux contribution is set by the corresponding left eigenvector l_i.

For a reversible chain the left and right eigenvectors are not independent but are related through the stationary distribution,

$$l_i = \Pi\, r_i, \qquad \Pi = \mathrm{diag}(\pi), \tag{B25}$$

that is, $l_i(x) = \pi_x\, r_i(x)$. The right eigenvectors are thus the equilibrium-population-weighted counterparts of the relaxation modes, and they are orthonormal in the π-weighted inner product,

$$\langle r_i, r_j\rangle_\pi := \sum_x \pi_x\, r_i(x)\, r_j(x) = \delta_{ij}. \tag{B26}$$

The sign structure of the dominant eigenvectors provides the geometric criterion used by PCCA++ [7, Ch. 2.5.1–2.5.2] to lump microstates into metastable macrostates. I use this to connect the slowest exchange modes of the Markov chain to the folded/unfolded thermodynamics of the protein, keeping in mind that spectral metastability and thermodynamic stability are related but not strictly identical notions.

Implied-timescale test and lag-time selection. Markovianity was assessed, and the lag time selected, through the implied-timescale (ITS) convergence test. The estimated timescales (B23) are variational lower bounds on the true timescales t_i^* of the underlying continuous-state process, $t_i(\tau) \leq t_i^*$, with the gap shrinking as the discretisation is refined or as τ is increased. In practice $t_i(\tau)$ is plotted against τ and one looks for a range over which the timescales are approximately constant; the lag time is taken as the smallest τ (counted in frames, $\tau = 1, 2, 3, \ldots$) inside this range, which minimises the loss of temporal resolution while keeping the process of interest approximately Markovian [7, Ch. 4.7]. Because the number of microstates and the lag time jointly control resolution and statistical accuracy, the two were chosen together (see main text, Sec. III A).